

CHAPTER – II

PROGRAMMING FUNDAMENTALS

- Data types
- Variables
- Type casting
- Operators in java
- Primitive data types
- Non-primitive data types
- Java syntax rules
- comments in java
- input and output in java
- Structure of java program

DATA TYPES:-

A data type defines what kind of value a variable can store and how much memory it will take.

- int,double,char,Boolean and string are data types
- Allocate proper memory for variables
- Operation can perform on

VARIABLES:-

Variables is a container in computer that store data or information. Variables is also known also attribute or data .

- **SYNTAX:-**

datatype variableName= value;

int age =25

TYPES OF VARIABLES:-

- Local variables
- Instance variables
- Static variables

RULES FOR NAMING VARIABLES:-

- Variable names cannot have space.
- Variable names cannot starts with a number
- Variable names cannot have java keyword like class,int,static.
- Variable name must start with a letter,\$,or_.

TYPE CASTING:-

Type casting means converting one data type into another.

IMPLICIT TYPE CASTING:-

- Done automatically by the compiler.
- Converts a smaller data type to a large data type.

EXPLICIT TYPE CASTING:-

- Done by the programmer using a cast operator.

- Converts a larger data type to a smaller data type.

OPERATORS IN JAVA:-

Operators are special symbols used to perform operations on variables and values. They are used to perform calculations, comparison, logical operations, assignment, and other tasks in java programs.

- Arithmetic operator
- Relational operator
- Logical operator
- Assignment operator
- Increment operator
- Decrement operator
- Bitwise operator
- Conditional operator
- Unary operator

PRIMITIVE DATA TYPES:-

Primitive datatype has 8 datatypes

- Int – 4 bytes
- Float – 4 bytes
- Long – 8 bytes
- Double - 8 bytes
- Short - 2 bytes
- Char – 2 bytes

- Boolean – 1 bit
- Byte – 1 byte

NON-PRIMITIVE DATA TYPES:-

Non – primitive datatype has a datatypes

- String
- Array
- Class
- Object
- Interface

JAVA SYNTAX RULES:-

- Every statement ends with a semicolon(;).
- Curly braces{} define blocks of code.
- Class names should start with an uppercase letter.
- Method and variable names should start with a lowercase letter.
- Java is casesensitive:Hello ! hello.
- The file name must match the public class name.

COMMENTS IN JAVA:-

Comments are used to make code more understandable. they are ignored by the compiler.

// single line comment

/*

Multi-line comment

Used for longer explanations

*/

INPUT AND OUTPUT IN JAVA:-

In java, input and output are commonly handled using the scanner class and to print the text or values.

```
System.out.println("hello students!");
```

```
System.out.print("welcome");
```

STRUCTURE OF JAVA PROGRAM:-

- Package declaration(groups related classes together).
- Import statement(import predefined java classes or packages).
- Class declaration(every java program must have at least one class).
- Main method(entry point of the program).
- Statements(contains the program logic).